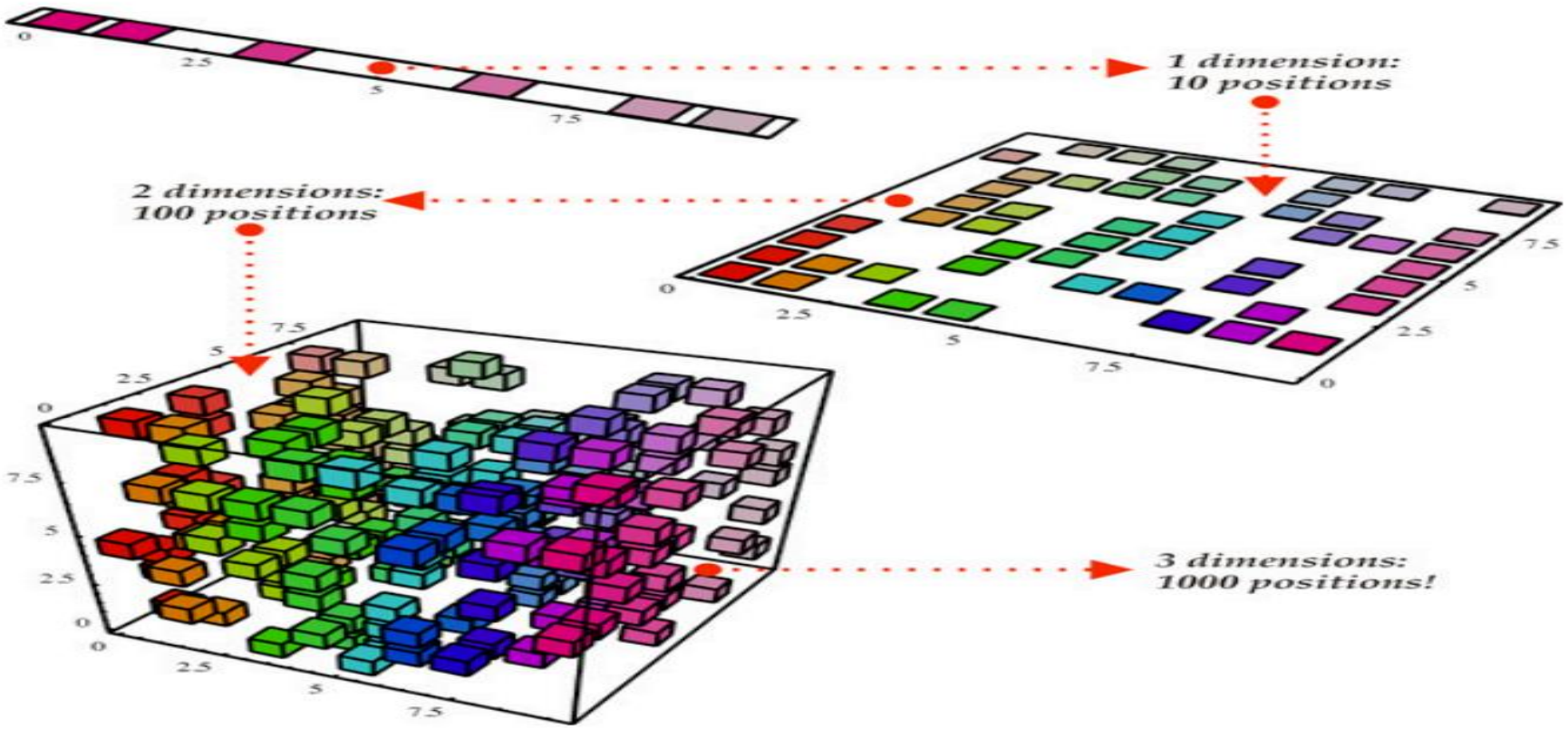


Chapter 6
Dimensionality Reduction

Chapter Outline

I. **Introduction**

II. **Decomposition algorithms**

- **Principal Component Analysis**
- **Kernel Principal Component Analysis**
- **Singular Value Decomposition**

III. Manifold learning algorithms

IV. Discriminant Analysis

I. Introduction

Why do you need dimensionality reduction?

Issues that arise with high dimensional data are:

1. Difficulty in clustering similar features.
2. Increased space and computational time complexity.
3. Running a risk of overfitting the machine learning model.
4. Unable to visualize/understand the data.

I. Introduction

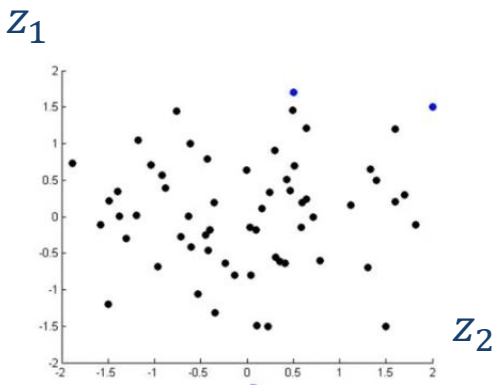
Example

- Number of features: 50

Country	GDP (trillions of US$)	Per capita GDP (thousands of intl. $)	Human Develop-ment Index	Life expectancy	Poverty Index (Gini as percentage)	Mean household income (thousands of US$)	...
Canada	1.577	39.17	0.908	80.7	32.6	67.293	...
China	5.878	7.54	0.687	73	46.9	10.22	...
India	1.632	3.41	0.547	64.7	36.8	0.735	...
Russia	1.48	19.84	0.755	65.5	39.9	0.72	...

- Reduced to 2 features

Country	z_1	z_2
Canada	1.6	1.2
China	1.7	0.3
India	1.6	0.2
Russia	1.4	0.5



I. Introduction

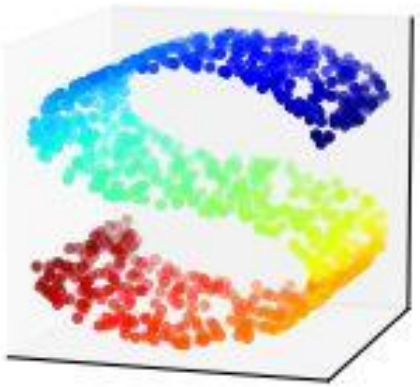
Decomposition algorithms	Principal Component Analysis	Unsupervised algorithms
	Kernel Principal Component Analysis	
	Singular Value Decomposition	
	Non-Negative Matrix Factorization	
Manifold learning algorithms	t-Distributed Stochastic Neighbor Embedding	
	Spectral Embedding	
	Locally Linear Embedding	
Discriminant Analysis	Linear Discriminant Analysis	Supervised algorithms
	Quadratic Discriminant Analysis	

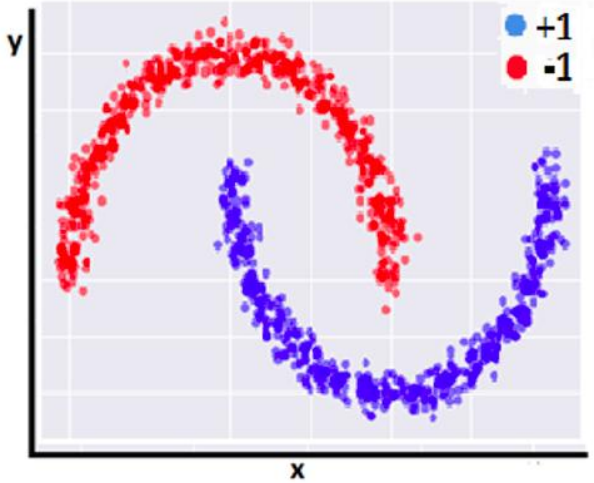
I. Introduction

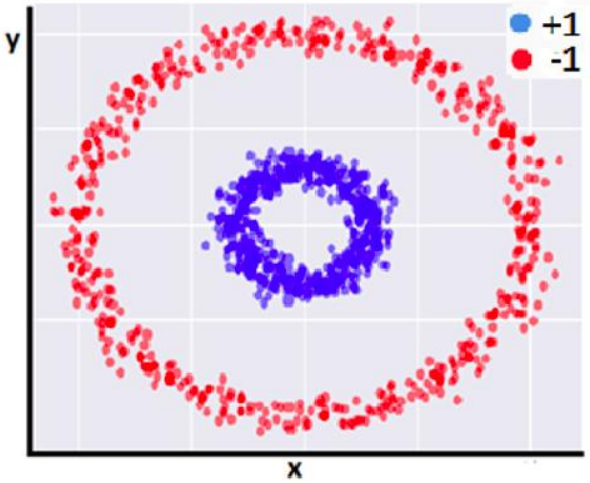
Non-linear distributions

- describe a situation where **there is not a straight-line (or linear relationships)** or direct relationship between an independent variable and a dependent variable.

- Dimensionality reduction techniques could be **categorized** between applied on linear and non-linear distributions



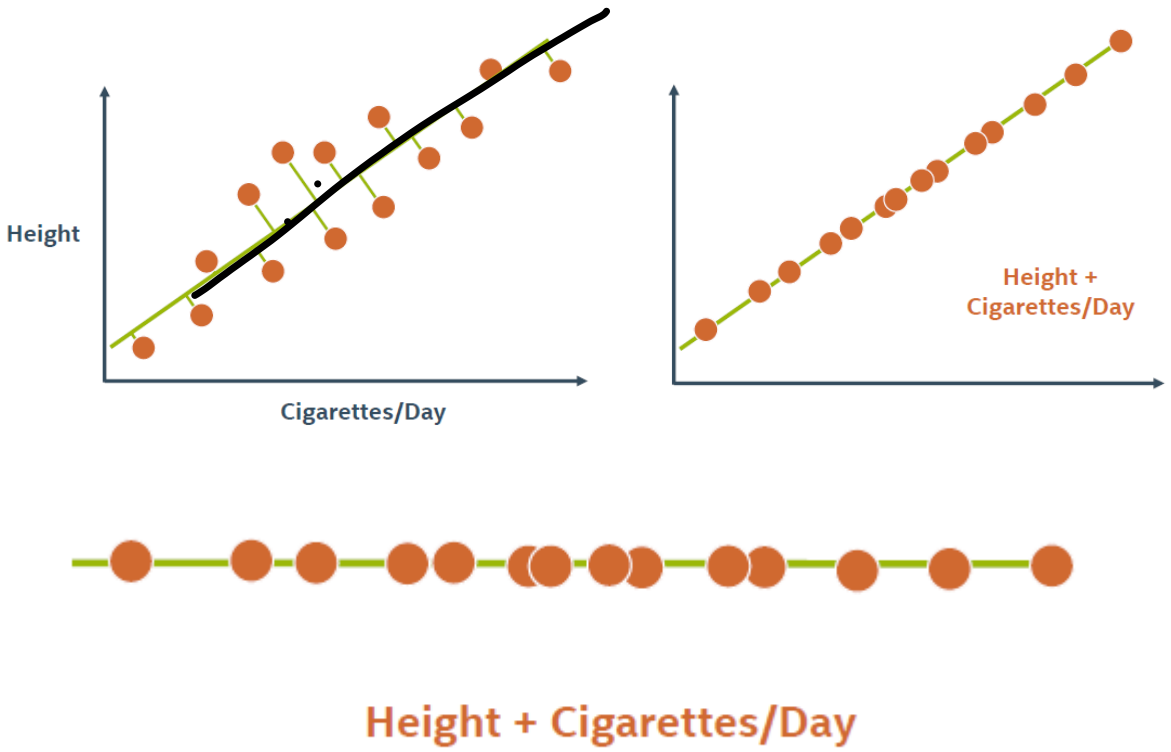




II. Decomposition Algorithms

PCA

- principal components are the new set of variables that are obtained from the initial set of variables.
- The principal components are computed in such a manner that newly **obtained variables are highly significant and independent of each other**.



II. Decomposition Algorithms

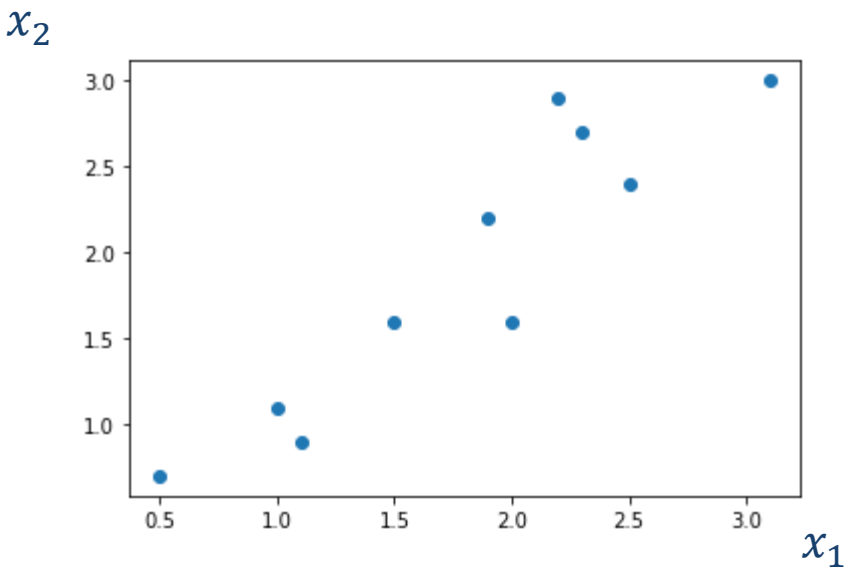
PCA - steps

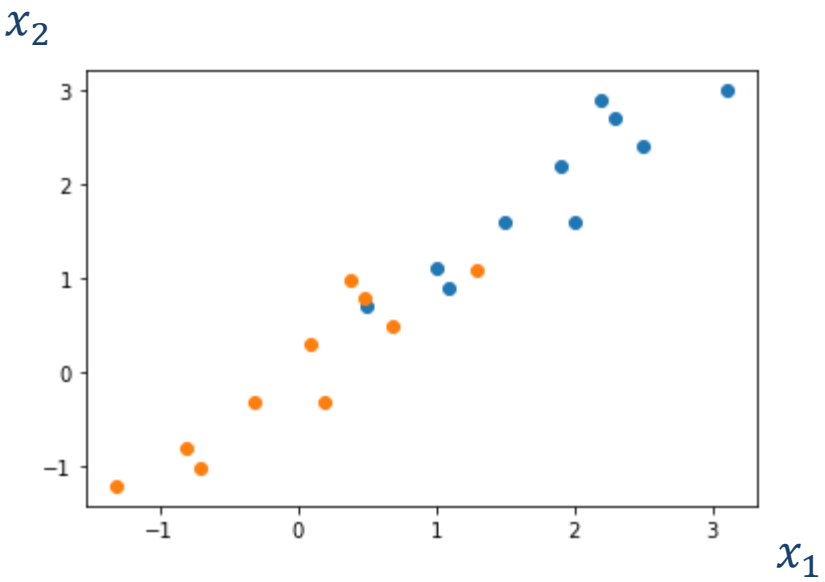
	Standardization of the data	Computing the covariance matrix	Calculating the eigenvectors and eigenvalues	Computing the Principal Components	Reducing the dimensions of the data set
Outputs	▪ Standardized data ▪ Mean = 0	▪ COV matrix (nxn) ▪ Relationships in the entire dimensions as the relationships between every 2 Random Variables	▪ Eigenvectors - axes (directions) ▪ Eigenvalues - radius of Ellipse	▪ the new features	• K components

II. Decomposition Algorithms

PCA – step 1 : standardisation of the data

you transform your data such that the resulting distribution has a mean of 0 and a standard deviation of 1.





II. Decomposition Algorithms

PCA – step 2 : Computing the covariance matrix

$$Cov(X,Y) = \frac{\sum (X_i - \bar{X}) * (Y_i - \bar{Y})}{n}$$

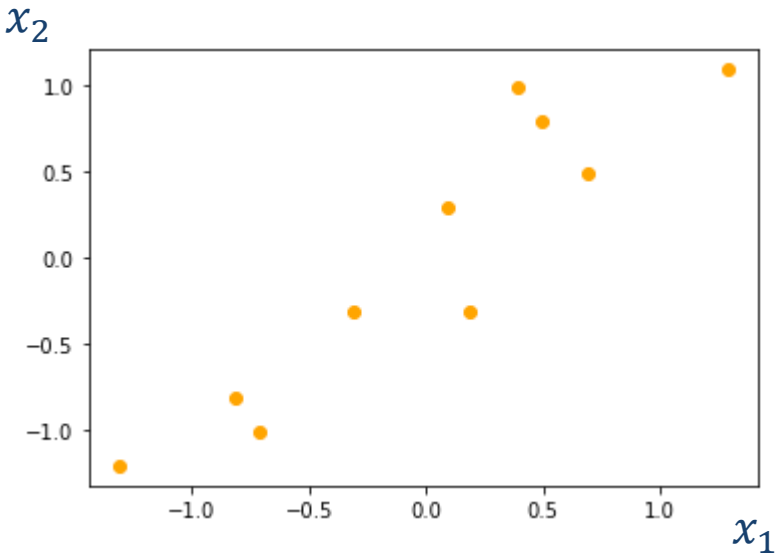
- covariance matrix is a n x n matrix, where n represents the dimensions of the dataset.
- The covariance value denotes how co-dependent two variables are with respect to each other
- If the covariance value is negative, it denotes the respective variables are indirectly proportional to each other
- A positive covariance denotes that the respective variables are directly proportional to each other
- **Some versions does not write the mean of X and Y since they assume the data is already centered on zero.**

II. Decomposition Algorithms

PCA – step 2 : Computing the covariance matrix

In our example:

- There is more variability in x_2 than x_1
- x_1 and x_2 covary in a positive way.

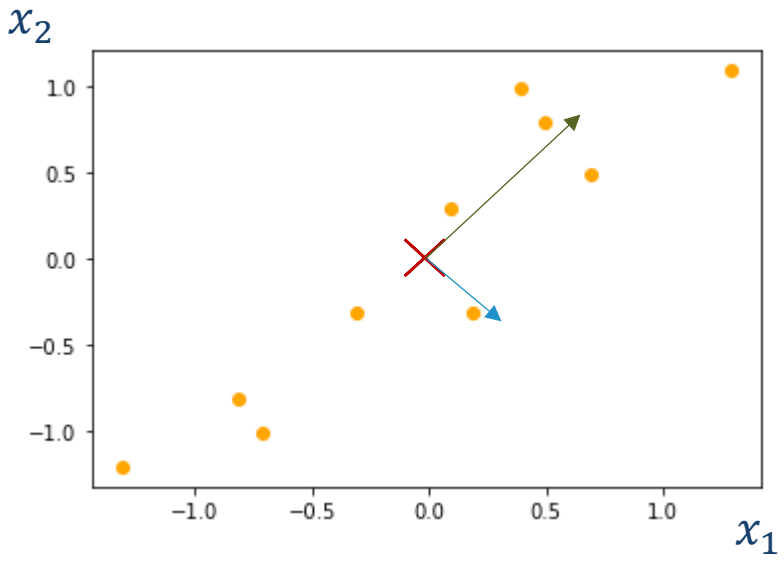


In other examples:

- You can deduct which variables tend to covary in a positive way, negative way or no covariance.

COV	x_1	x_2
x_1	0.616556	0.615444
x_2	0.615444	0.716556

PCA – step 3 : Calculating the Eigen Vector and Eigen Values



Variable	Eigenvector 1	Eigenvector 2
x_1 x_2	$\begin{pmatrix} 0.67 \\ 0.73 \end{pmatrix}$	$\begin{pmatrix} 0.73 \\ -0.67 \end{pmatrix}$
EigenValues	1.284	0.049
%of total variance	1.28 / 1.33 = 96.3 %	0.049/1.33 = 3.7 %

COV	x_1	x_2
x_1	0.616556	0.615444
x_2	0.615444	0.716556

You can check the calculations in this link
Note that, they are calculated from the COV matrix

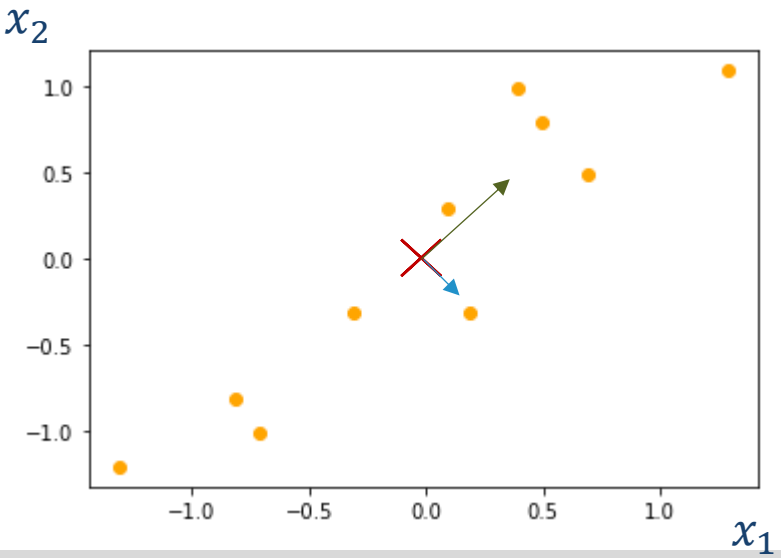
II. Decomposition Algorithms

- Eigen values determine the radius of the Ellipse
- for every eigenvector there is an eigenvalue. The dimensions in the data determine the number of eigenvectors that you need to calculate.
- The idea behind eigenvectors is to use the Covariance matrix to understand where in the data there is the most amount of variance.

II. Decomposition Algorithms

PCA – step 4 : Calculating the Principal components

The final step in computing the Principal Components is to form a matrix known as the feature matrix that contains all the significant data variables that possess maximum information about the data.



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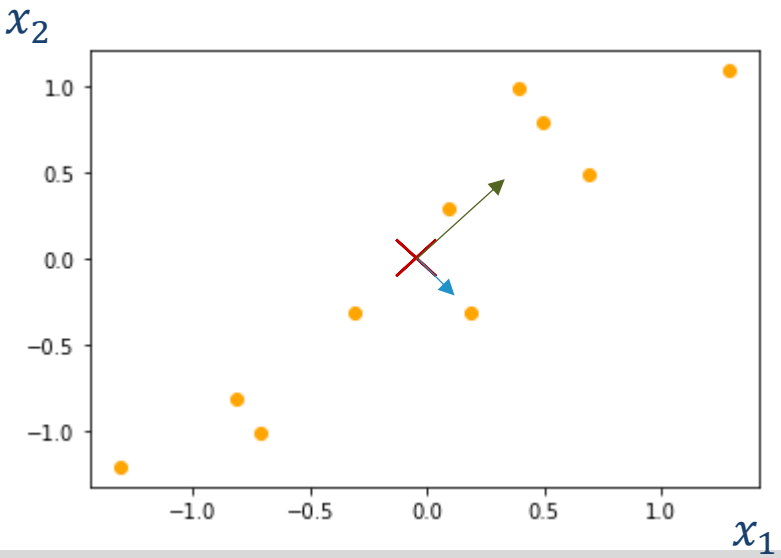
II. Decomposition Algorithms

PCA – step 4 : Calculating the Principal components

y* = *W'* × *x where
W *is the eigenvector matrix*
W' is the *transpose* of the matrix ***W***

The final step in computing the Principal Components is to form a matrix known as the feature matrix that contains all the significant data variables that possess maximum information about the data.



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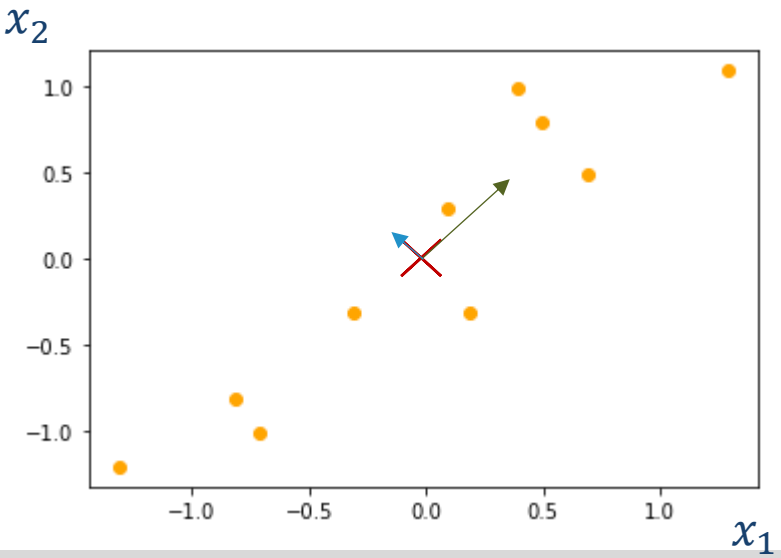
%of total variance	1.28 / 1.33 = 96.3 %	0.049/1.33 = 3.7 %

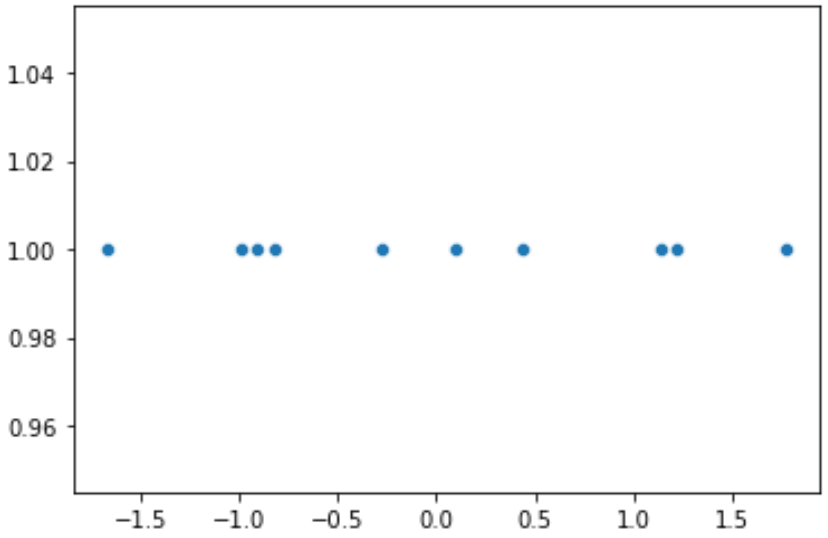
II. Decomposition Algorithms

PCA – step 5 : Choose k

- Project the feature space via PCA onto a smaller subspace, where the eigenvectors will form the axes of this new feature subspace.

- You can also calculates how much the data variance is retained





PCA – step 5 : Choose k

- How to choose k ?
  - **For visualization k = 2**
  - **Typically, we want the explained variance to be between 95–99%. In python you can specify the components = 0.95 instead of the number of dimensions**
  - **Otherwise: When the number of explained variation converge.**

II. Decomposition Algorithms

PCA implementation

```
def pca(X=np.array([]), no_dims=50):
print("Preprocessing the data using PCA...")
(n, d) = X.shape
Mean = np.tile(np.mean(X, 0), (n, 1))
X = X - Mean
(l, M) = np.linalg.eig(np.dot(X.T, X))
Y = np.dot(X, M[:, 0:no_dims])
return Y
```

II. Decomposition Algorithms

- PCA in the data science process
  - Apply on training set (fit)
  - Predict the testing set (transform)
  - Check the maintained variance on the testing data.

- PCA in more dynamic way
  - Apply the PCA on Train data with certain variance percentage
  - Apply your model (classification, regression, clustering, …)
  - Transform the testing set
  - Apply the prediction.

II. Decomposition Algorithms

PCA summary

- It is a dimensionality-reduction method to **find lower-dimensional** space by preserving the **variance** as measured in the high dimensional input space.
- It is an **unsupervised** method for dimensionality reduction.
- PCA transformations are linear transformations. It involves the process of finding the principal components, which is the decomposition of the feature matrix into eigenvectors. This means that PCA will not be effective when the distribution of the dataset is non-linear.

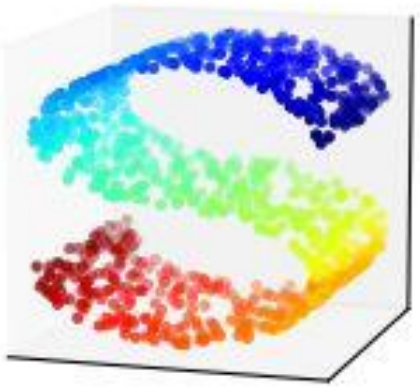
II. Decomposition Algorithms

Kernel PCA

- The PCA transformations we described previously are linear transformations that are ineffective with the non-linear distribution. To deal with non-linear distribution, the basic idea is to use the **kernel trick**.
- **A kernel trick**
  - a method to project non-linear data onto a higher dimensional space and separate different distributions of data.
  - Once the distributions are separated we can use PCA to separate them linearly.

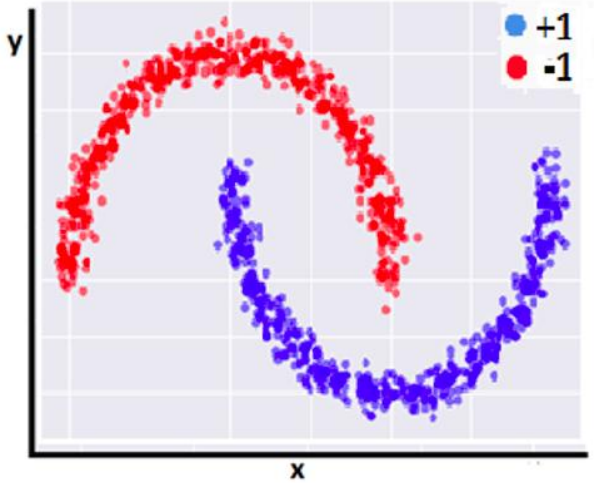
II. Decomposition Algorithms

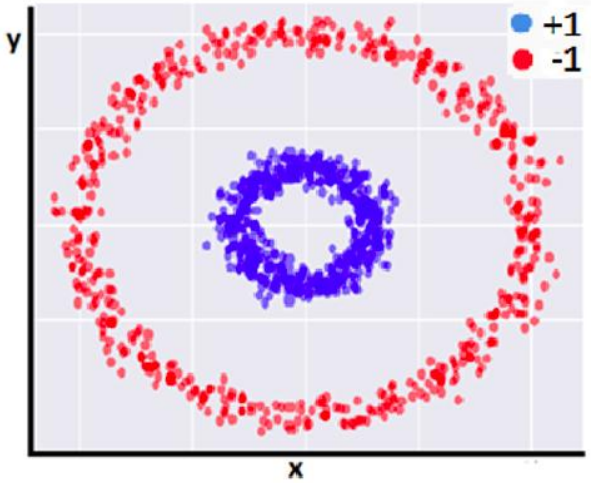
Kernel PCA



Non-linearity distributions

describe a situation where **there is not a straight-line** or direct relationship between an independent variable and a dependent variable.

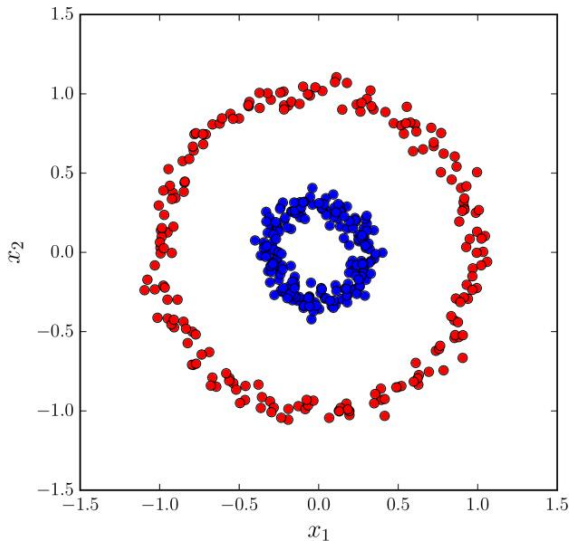


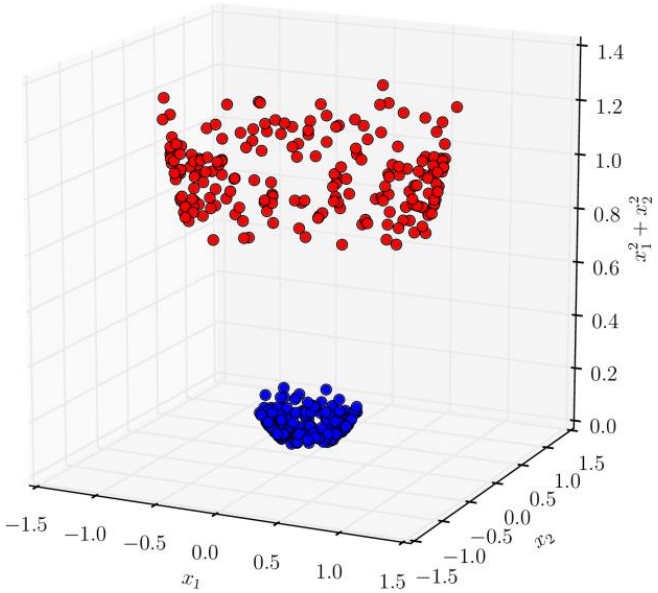


II. Decomposition Algorithms

Kernel PCA

- you will try to increase the dimensions, then apply the PCA.
- if we will introduce new feature $z_1^2 + z_2^2$ and extend our space of features we will get the following picture:



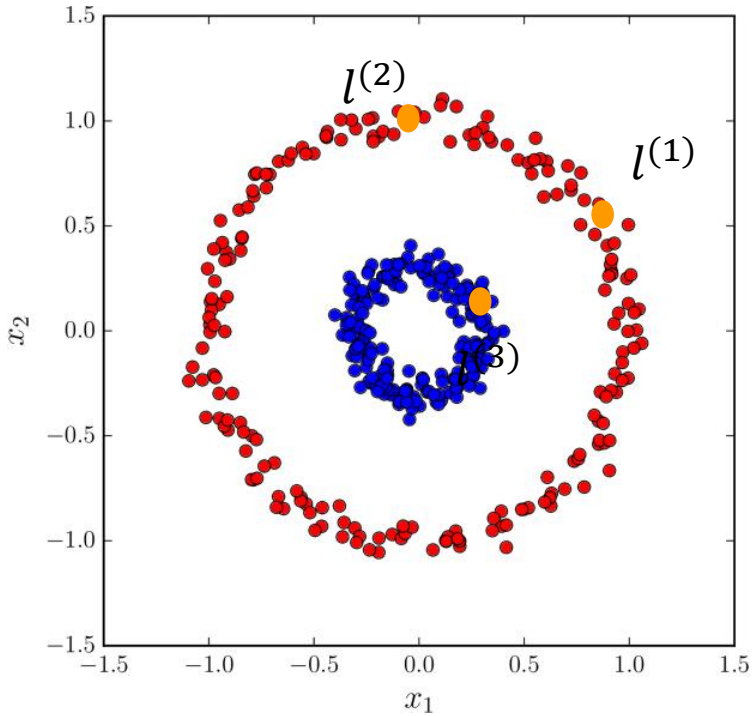


II. Decomposition Algorithms

Kernel PCA

How to create new features?

- Consider three points : $l^{(1)}, \ l^{(2)}, \ l^{(3)}$
- Given x, compute new feature depending on proximity to landmarks: $l^{(1)}, \ l^{(2)}, \ l^{(3)}$



- For each point, 3 new features are generated
  - $f_1 = K\left(x, l^{(1)}\right) = \exp\left(-\frac{\left\|x - l^{(1)}\right\|^2}{2\sigma^2}\right)$
  - $f_2 = K\left(x, l^{(2)}\right) = \exp\left(-\frac{\left\|x - l^{(2)}\right\|^2}{2\sigma^2}\right)$
  - $f_3 = K\left(x, l^{(3)}\right) = \exp\left(-\frac{\left\|x - l^{(3)}\right\|^2}{2\sigma^2}\right)$

The function K is also called Gaussian Kernel

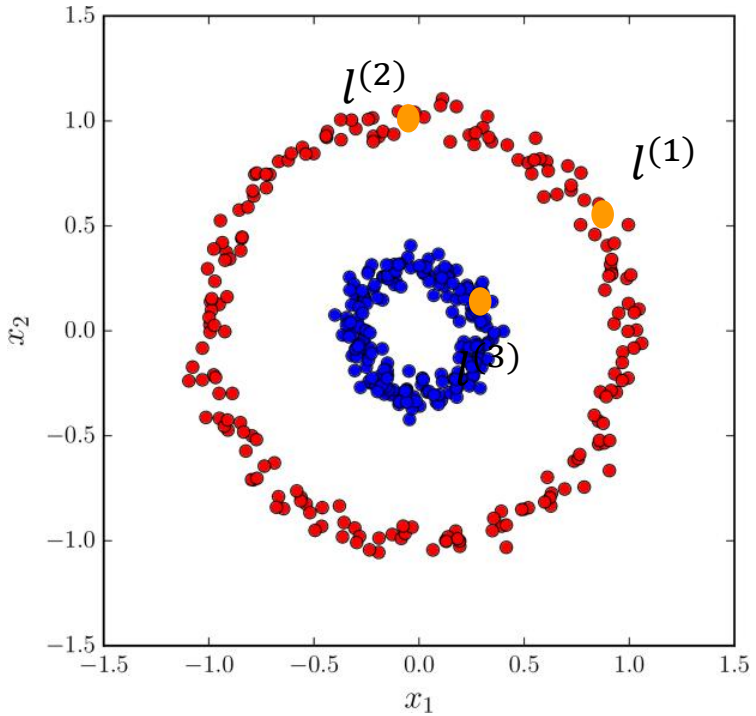
II. Decomposition Algorithms

Kernel PCA

What is the K method ?

- $f_i = K\left(x, l^{(i)}\right) = \exp\left(-\frac{\lambda\left\|x - l^{(i)}\right\|^2}{2\sigma^2}\right)$

- The K is a kernel method
- implicitly map objects to high-dimensional feature spaces
- If x close to lemma -> similarity ~ 1
- If x far from lemma -> similarity ~ 0



II. Decomposition Algorithms

Kernel PCA

- The RBF kernel function for two points X_1 and X_2 computes the similarity or how close they are to each other. This kernel can be mathematically represented as follows:
  - $k\,(X_1\,, X_2) = \exp(-\ \frac{||X_1 - X_2||^2}{2\sigma^2})$
  - Where:
    1. 'σ' is the variance and our hyperparameter
    2. $||X_1\ - X_2||^2$ is the Euclidean (L_2-norm) Distance between two points X_1 and X_2

- When **the points are the same**, the points are extremely similar and the distance is close to 1
- When **the points are separated by a large distance**, then the kernel value is close to 0 which would mean that the points are dissimilar

II. Decomposition Algorithms

Kernel PCA

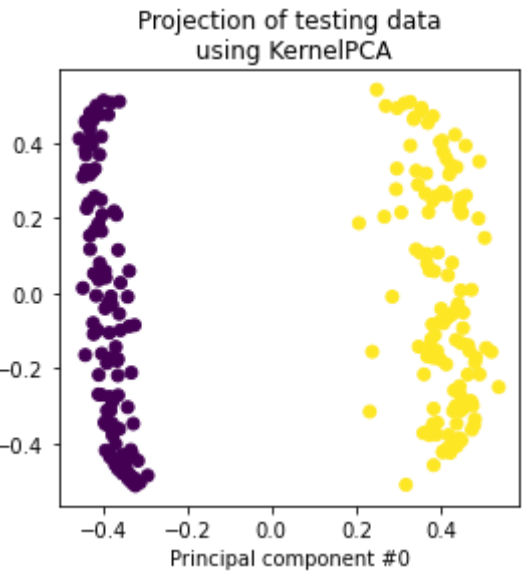
What is the number of landmarks ?

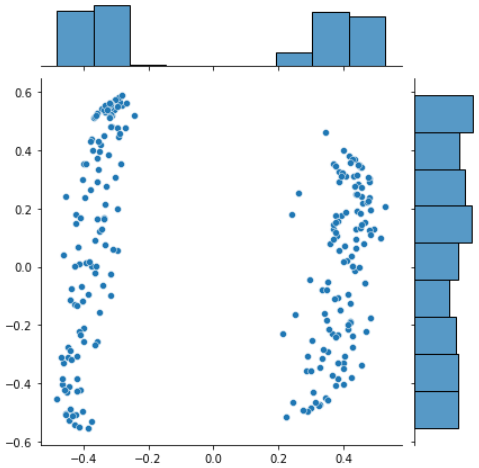
- Each point is a landmark.
- You will have data of size = sample size
- Choose the number of components you desire.

Kernel PCA - the lambda parameter λ

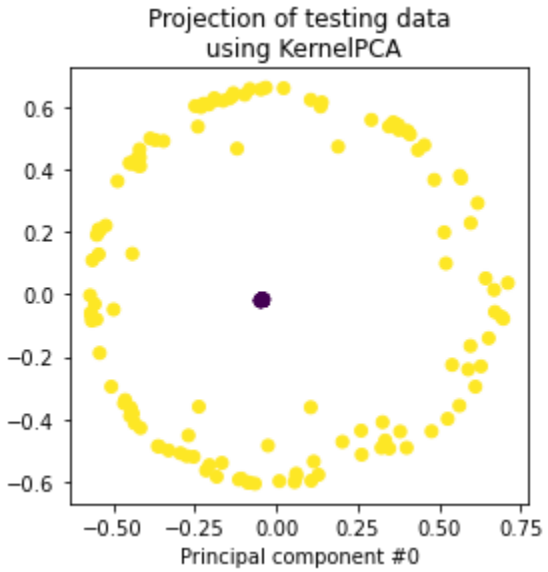
- Low Lambda -> flat bell

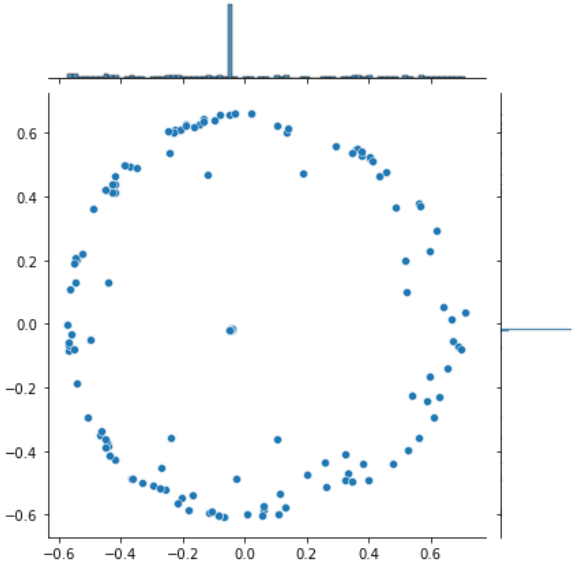
Gamma = 2





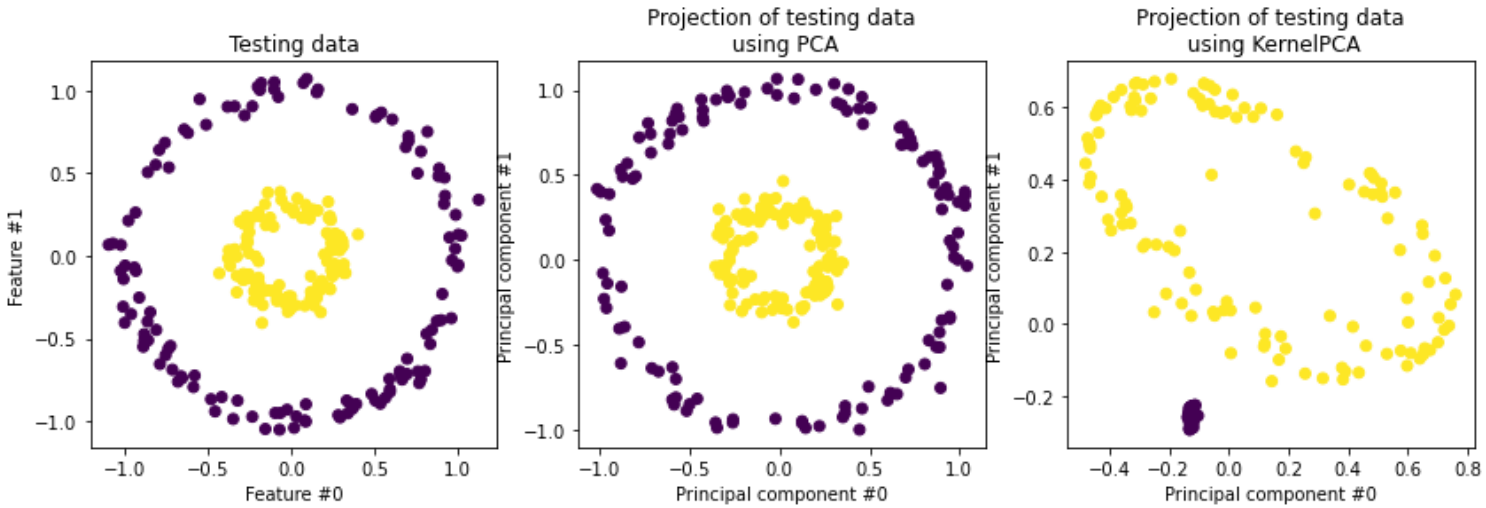
Gamma = 15

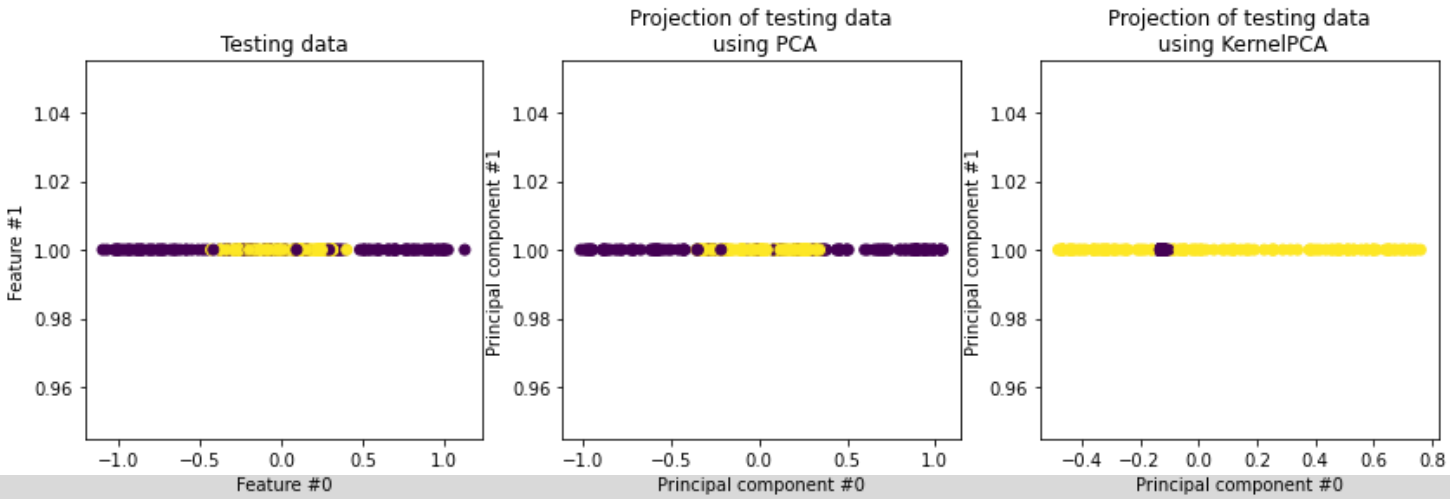




II. Decomposition Algorithms

- **After applying Kernel :**
  - You can apply PCA
  - Even though now it gave bad result when applied on 1D, we can improve it using LDA
  - The idea is, you achieved a separation that is linear

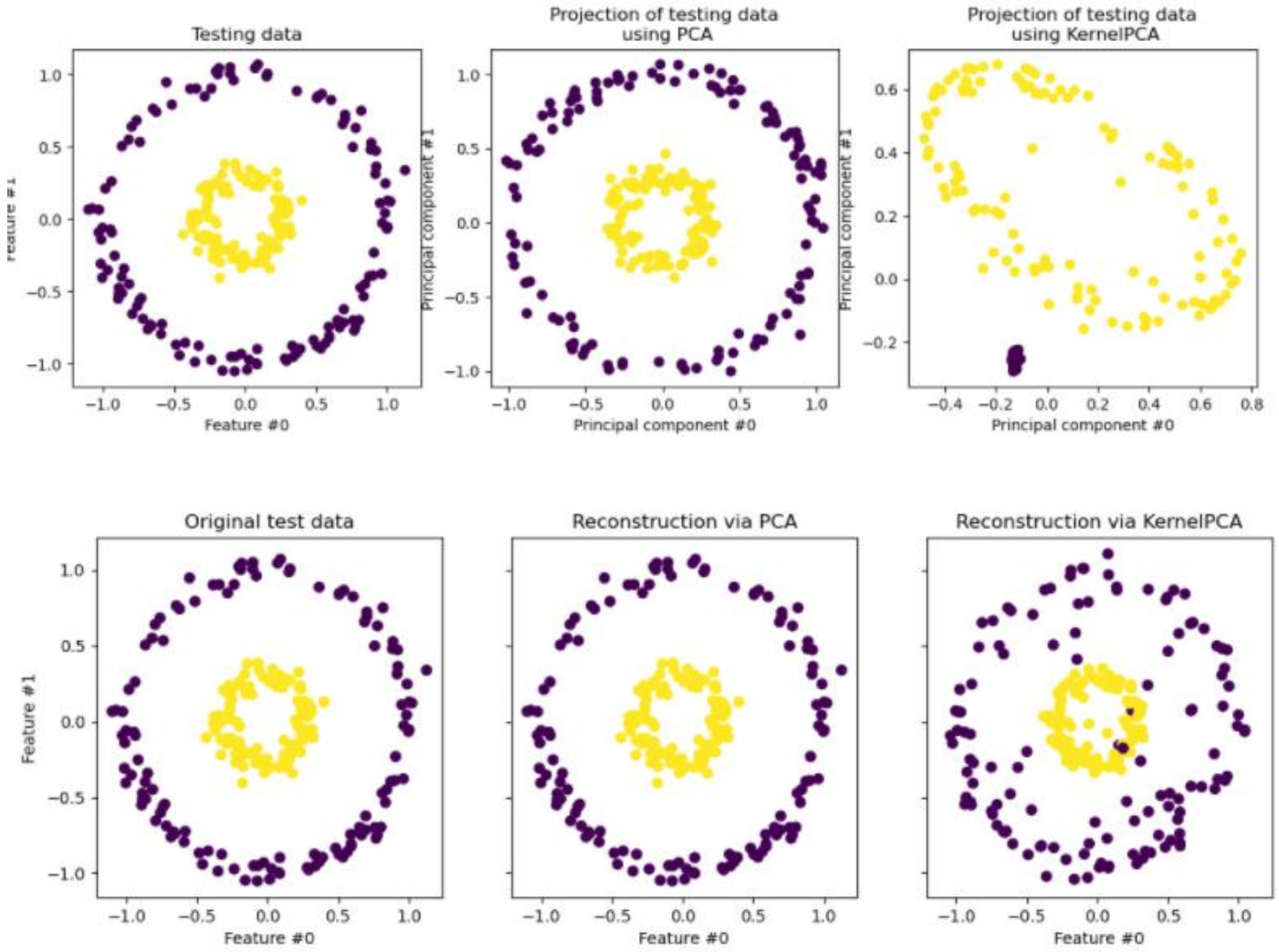




II. Decomposition Algorithms

➢ Reconstruction

Reverse transform will reconstruct the original data.



Given five samples $S1, S2, S3, S4, S5$ Each sample has three features x_1, x_2, x_3
Reduce the three features into two using PCA

$$\Rightarrow \text{Mean} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Features	S_1	S_2	S_3	S_4	S_5
x_1	$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$,	$\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$,	$\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$,	$\begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$,	$\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$
x_2					
x_3					

[1] Calculate Covarinace of all features

$$cov_X = \frac{1}{N}\sum_{i=1}^{N}(\underline{X}_i - \overline{\underline{X}}_i)(\underline{X}_i - \overline{\underline{X}}_i)^T$$

Since samples S_1, S_2, s_3, S_4, S_5 are zero mean $\Rightarrow$ Use Samples directly for Covariance calculations

$$\textbf{Covariance} = \frac{1}{5}\left[\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 1 & 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}\begin{pmatrix} -1 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}\begin{pmatrix} 1 & -1 & -1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}\begin{pmatrix} -1 & -1 & -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}\begin{pmatrix} 0 & 0 & 2 \end{pmatrix}\right] =$$

$$= \frac{1}{5}\begin{bmatrix} 1+1+1+1+0 & 1-1+1+1+0 & 0+0-1+1+0 \\ 1-1-1+1+0 & 1+1+1+1+0 & 0+0+1+1+0 \\ 0+0-1+1+0 & 0+0+1+1+0 & 0+0+1+1+4 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 2 \\ 0 & 2 & 6 \end{bmatrix} = \begin{bmatrix} 0.8 & 0 & 0 \\ 0 & 0.8 & 0.4 \\ 0 & 0.4 & 1.2 \end{bmatrix}$$

[2] Calculate Eignevalues of covariance matrix: λ^*

$$\begin{vmatrix} 0.8-\lambda^* & 0 & 0 \\ 0 & 0.8-\lambda^* & 0.4 \\ 0 & 0.4 & 1.2-\lambda^* \end{vmatrix} = 0 \quad \xrightarrow[\Rightarrow]{\text{multiply by 5}} \quad \frac{1}{5}\begin{vmatrix} 4-5\lambda^* & 0 & 0 \\ 0 & 4-5\lambda^* & 2 \\ 0 & 2 & 6-5\lambda^* \end{vmatrix} = 0$$

To Simplify the calculations, let $\lambda = 5\lambda^*$ *At the end, divide calculated* λ *by* 5

$$\frac{1}{5}\begin{vmatrix} 4-\lambda & 0 & 0 \\ 0 & 4-\lambda & 2 \\ 0 & 2 & 6-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)[(4-\lambda)(6-\lambda)-4]=0$$
$$(4-\lambda)[24+\lambda^2-10\lambda-4]=0$$
$$(4-\lambda)[\lambda^2-10\lambda+20]=0$$

$$Eq: ax^2+bx+C$$
$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$\lambda_1 = \mathbf{4}$ $\quad \lambda_2 = \mathbf{5}+\sqrt{\mathbf{5}} = 7.24$ $\quad \lambda_3 = \mathbf{5}-\sqrt{\mathbf{5}} = 2.76$

Remember: Actual λ are those values divided by 5

$\lambda_1^* = \mathbf{4}/5 = 0.8$ $\quad \lambda_2^* = \mathbf{7.24}/5 = 1.448$ $\quad \lambda_3^* = \mathbf{2.76}/5 = 0.553$

$$\frac{1}{5}\begin{pmatrix} 4-\lambda & 0 & 0 \\ 0 & 4-\lambda & 2 \\ 0 & 2 & 6-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

For $\lambda_1 = 4$

$$\begin{pmatrix} 4-4 & 0 & 0 \\ 0 & 4-4 & 2 \\ 0 & 2 & 6-4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow$$

From R_2: $x_3 = 0$

From R_3: $x_2 = 0$

From C_1: x_1 take any value (e.g. 1)

$$X_{\lambda=4} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Length = 1

For $\lambda_2 = 5+\sqrt{5}$

$$\begin{pmatrix} 4-(5+\sqrt{5}) & 0 & 0 \\ 0 & 4-(5+\sqrt{5}) & 2 \\ 0 & 2 & 6-(5+\sqrt{5}) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} (-1-\sqrt{5}) & 0 & 0 \\ 0 & (-1-\sqrt{5}) & 2 \\ 0 & 2 & (1-\sqrt{5}) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow$$

From R_1: $x_1 = 0$

$$\begin{array}{r} \\ R_2*(1-\sqrt{5}) \\ R_3*2 \end{array}\begin{pmatrix} (-1-\sqrt{5}) & 0 & 0 \\ 0 & (-1-\sqrt{5})(1-\sqrt{5}) & 2(1-\sqrt{5}) \\ 0 & 2*2 & 2(1-\sqrt{5}) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \qquad \begin{array}{r} \\ R_2-R_3 \\ \\ \end{array}\begin{pmatrix} (-1-\sqrt{5}) & 0 & 0 \\ 0 & 4 & 2(1-\sqrt{5}) \\ 0 & 4 & 2(1-\sqrt{5}) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{pmatrix} (-1-\sqrt{5}) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 4 & 2(1-\sqrt{5}) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

from R_3: $4x_2 + 2(1-\sqrt{5})x_3 = 0$

$x_2 = -1/2(1-\sqrt{5})x_3$

Let $x_3 = 2$ then $x_2 = -(1-\sqrt{5})$

$\Rightarrow$

$$X_{\lambda=5+\sqrt{5}} = \begin{pmatrix} 0 \\ -(1-\sqrt{5}) \\ 2 \end{pmatrix}$$

$$Length = \sqrt{0+[-(1-\sqrt{5})]^2+2^2}$$

$$= \sqrt{(10-2\sqrt{5})}$$